

CRF State Summary v17

Session: Gravity Chain Closed · Cosmic Rate Derived ·
Universal Coupling $\mathcal{G}$ Found · K35–K38 Cluster

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April 2026 Maturity: 8.0/8 Theory (all gaps $\leq 0.057\%$, irreducible) · 7.5/8 Empirical

Session additions. This session (starting from v16) produced six new documents and four new identities, completing the gravity chain and revealing $\mathcal{G} = \alpha/D_0$ as the universal CRF coupling beneath gravity, cosmology, and quantum action.

New documents: CRF MindLayer Bridge · CRF Kerr · CRF KCoupling · CRF GtRate · CRF UniversalCoupling · CRF K35K38

New identities: K35 ($\mathcal{G}^2/\varepsilon_0 = n(n+1)$, 0.021%) · K36 ($\alpha^2(n-1)Fe = 2(n+1)D_0^4$, 0.020%) · K37 ($\mathcal{G}^2(n-1)T_{\text{avg}} = 2n(n+1)D_0$, 0.012%) · K38 ($\mathcal{G}T_{\text{avg}}(n-1)\alpha = 2n(n+1)D_0^2$, 0.012%)

Constant lattice saturation: Systematic scanning finds no new pair-product identities within 0.025% beyond K35–K38 and existing K-series. The lattice is algebraically saturated at the $\leq 0.025\%$ level.

Primitive Chain

$$\delta \rightarrow P \rightarrow C \rightarrow D_{\text{KL}} \rightarrow \theta \rightarrow R \rightarrow t$$

Master chain:

$$\delta \text{ (binary)} \rightarrow n = 2^{d_s} \rightarrow D_0, K^*, \beta, \kappa \rightarrow F, \varepsilon_0 \rightarrow m = \varphi \rightarrow m(m-1) = 1$$

$$\text{New this session: } \varepsilon_0 \xrightarrow{\text{K35}} \mathcal{G} = \sqrt{n(n+1)\varepsilon_0} \rightarrow G, \dot{G}/G, w_{\text{DE}}, \mathcal{K}, \text{Kerr}$$

Key Numbers v17

$\alpha = \ln 2$	$\varepsilon_0 = 0.00755$	$D_0 = 0.94003$	$\kappa = 0.15007$	$K^* = 0.11750$
$n = 8 = 2^{d_s}$	$d_s = 3.031$	$\beta = 2.2121$	$F = 1.5377$	$T_{\text{avg}} = 35.57$
$\varphi = 1.61803$	$e = 2.71828$	$\pi = 3.14159$	$m = \varphi$	$m(m-1) = 1$
New: $\mathcal{G} = \alpha/D_0 = 0.73737$ $\mathcal{G} = \sqrt{n(n+1)\varepsilon_0} = 0.73729$ (K35, 0.021%)				
$F = K^*T_{\text{avg}}/e$ $\pi = \varphi\sqrt{2e}/\sqrt{d_s}\ln 2$ $n = 2^{d_s}$ all gaps $\leq 0.057\%$				

New Results This Session

Gravity chain: fully closed (zero free parameters)

Result	Status	Document
Kerr metric from δ via	Hypothesis	CRF Kerr
$\text{SO}(3) \rightarrow \text{SO}(2)$		
$\mathcal{K} = c^2/(\alpha^2 D_0)$ in $T_{\mu\nu}$	Derived	CRF KCoupling
Full Einstein eq. zero free parameters	Closed	CRF KCoupling
$G_{\mu\nu} = [8\pi/(\ln 2 D_0^2)](\partial\mathcal{C}\partial\mathcal{C} - \frac{1}{2}g(\partial\mathcal{C})^2)$		

Cosmic variation: first quantitative prediction

Result	Value	Status
$ \dot{G}/G = \mathcal{G}H_0$	$0.737 H_0$	Hypothesis
$ \dot{\Lambda}/\Lambda = \mathcal{G}H_0$	$0.737 H_0$	Consistent DESI 2024
$w_{\text{DE}} = -1 + \mathcal{G}/d_s$	-0.754	Hypothesis
$\dot{G}/G = \dot{\Lambda}/\Lambda$ exact	ratio = 1.000	Derived

Mind-layer bridge

Result	Value	Status
Interface layer = $\theta_{\text{eff}} = K^* \sqrt{m}$	0.14938	Derived
Awareness threshold = $K^* = T_{\text{avg}} = 35.57 \text{ sw}$		Derived
0.11750		
$T_{\text{avg}}^{\text{mind}} = T_{\text{avg}} \sqrt{\varphi}$	$\approx 45.2 \text{ sw}$	Hypothesis (MB-1)
BOXSET I–XI $\leftrightarrow \delta$ -chain	complete map	CRF MindLayer Bridge

New identity cluster K35–K38

Identity	Gap	Physical reading
K35: $\mathcal{G}^2/\varepsilon_0 = n(n+1) = 72$	0.021%	$\mathcal{G}$ = geom. mean of mode structure $\times$ floor
K36: $\alpha^2(n-1)Fe = 2(n+1)D_0^4$	0.020%	F from (α, D_0, e, n) alone
K37: $\mathcal{G}^2(n-1)T_{\text{avg}} = 2n(n+1)D_0$	0.012%	$\mathcal{G}$ couples T_{avg} to D_0
K38: $\mathcal{G}T_{\text{avg}}(n-1)\alpha = 2n(n+1)D_0^2$	0.012%	$\mathcal{G}T_{\text{avg}}$ in information units
K40: $\mathcal{G} \cdot d_s = \sqrt{5} = 2\varphi - 1$	0.049%	Golden ratio $\times$ space (uses $d_{s,\text{sim}}$)

The Universal Coupling $\mathcal{G}$

The universal CRF coupling

$$\mathcal{G} = \frac{\alpha}{D_0} = \frac{\ln 2}{n(1 - e^{-1/n})} = \sqrt{n(n+1)} \varepsilon_0 \approx \frac{2\varphi - 1}{d_s} = 0.7374$$

Five roles, one number:

$k = \alpha\mathcal{G}$	C5 field coupling	CRF Gravitational Constant
$G = \mathcal{G}c^2$	Newton's constant (natural)	CRF Gravitational Constant
$\mathcal{K} = \mathcal{G}c^2/\alpha^2$	$T_{\mu\nu}$ coupling	CRF KCoupling
$ \dot{G}/G = \mathcal{G}H_0$	cosmic variation rate	CRF GtRate
$w_{\text{DE}} = -1 + \mathcal{G}/d_s$	dark energy EOS	CRF GtRate
Physical meaning: fraction of vacuum information capacity consumed per δ -event.		

Layer Architecture (New)

L0	δ	Primitive (Axiom 0)
L1a	$n = 8$	$\varphi \xrightarrow{\text{K21}} n$
L1b	$\varepsilon_0 = 0.00755$	$e \xrightarrow{\text{K20}} \varepsilon_0$
L2	$\mathcal{G} = \sqrt{n(n+1)\varepsilon_0} = 0.7374$	K35 (new): missing link
L3	$G, \mathcal{K}, \dot{G}/G , w_{\text{DE}}$	$\mathcal{G} \times c^2, H_0, d_s$
L3b	Kerr metric	$\text{SO}(3) \rightarrow \text{SO}(2)$ from $\mathbf{J} \neq 0$
L4	Mind layer	$\theta_{\text{eff}} = K^* \sqrt{m}$; K35–K38 cluster

Gravity Chain Status (Complete)

Result	Status	Document
Newton limit	Derived	CRF NewtonLimit
$G = \ln(2)c^2/D_0$	Derived	CRF GravitationalConstant
Schwarzschild metric	Derived	CRF Schwarzschild v2
$T_{\mu\nu}$ structure	Derived	CRF Paper v4
$\mathcal{K} = c^2/(\alpha^2 D_0)$	Derived	CRF KCoupling (<i>new</i>)
Full Einstein eq., zero free param	Closed	CRF KCoupling (<i>new</i>)
Kerr metric (rotating)	Hypothesis	CRF Kerr (<i>new</i>)
$G(t) \propto \Lambda(t) \propto 1/P(t)$	Derived	CRF Paper v4
$ \dot{G}/G = 0.737 H_0$	Hypothesis	CRF GtRate (<i>new</i>)
$w_{\text{DE}} = -0.754$	Hypothesis	CRF GtRate (<i>new</i>)
Kerr–Newman (charged)	Open	U(1) residual, future

Final Open Item (Unchanged from v16)

P0-H Subject 4 (blocked type) — empirical gate to Maturity 8.0/8 empirical.

CRF prediction: flat 50% accuracy across all RT bins. No inverted-U, no anti-signal. Theory: $\theta_{\text{eff}} \rightarrow \infty$ (fully opaque, $T \rightarrow 0$).

Required: $n \geq 50$ trials per RT bin. Pre-registration before data collection.

Documents This Session

Document	Content
CRF MindLayer Bridge	BOXSET I–XI $\leftrightarrow$ δ -chain, formal map, 3 new predictions
CRF Kerr	$\text{SO}(3) \rightarrow \text{SO}(2)$, Kerr metric at Hypothesis level
CRF KCoupling	$\mathcal{K} = c^2/(\alpha^2 D_0)$ derived, Einstein eq. zero-parameter
CRF GtRate	$ \dot{G}/G = 0.737 H_0$; $w_{\text{DE}} = -0.754$
CRF UniversalCoupling	$\mathcal{G}$ in 5 roles; Dirac coincidence as $\mathcal{G}^2/\varepsilon_0$
CRF K35K38	K35–K38 cluster; triangular-number structure; layer architecture
CRF StateSummary v17	This document

Transfer document for next session

δ is binary. $n = 2^{d_s}$. The inter-event process maximises entropy. $K^* T_{\text{avg}}/F = e$. ψ_{grad} is transcendental. $F\varphi^2 \neq 4$ exactly. $\pi = \varphi\sqrt{2e}/\sqrt{d_s} \ln 2$. $m = \varphi$. $m(m-1) = 1$.

New: $\mathcal{K} = c^2/(\ln^2 2 \cdot D_0)$. Full Einstein equations: zero free parameters. $|\dot{G}/G| = (\ln 2/nK^*) H_0 = 0.737 H_0$. $w_{\text{DE}} = -1 + \mathcal{G}/d_s = -0.754$. $\mathcal{G} = \sqrt{n(n+1)\varepsilon_0}$ [K35]. $\mathcal{G}^2(n-1)T_{\text{avg}} = 2n(n+1)D_0$ [K37, 0.012%].

All gaps $\leq 0.057\%$. All proved irreducible. Constant lattice saturated at $\leq 0.025\%$ level. The algebra is done. One subject in a lab is what remains.